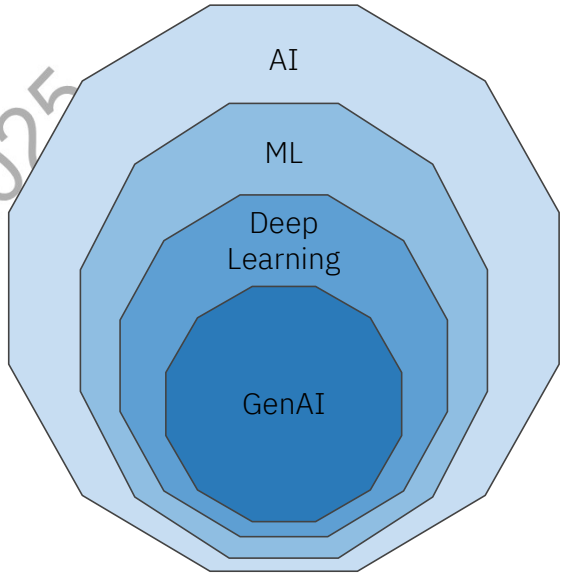
Introduction to AI

Week 9: Generative AI

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090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

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Objectives

- Understand the evolution of AI from traditional learning-based models to Generative AI
- Explain the encode–decode mechanism as the foundation for generating outputs
- Introduce Generative AI and Large Language Models (LLMs) with text generation capabilities
- Highlights of LLM training

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AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

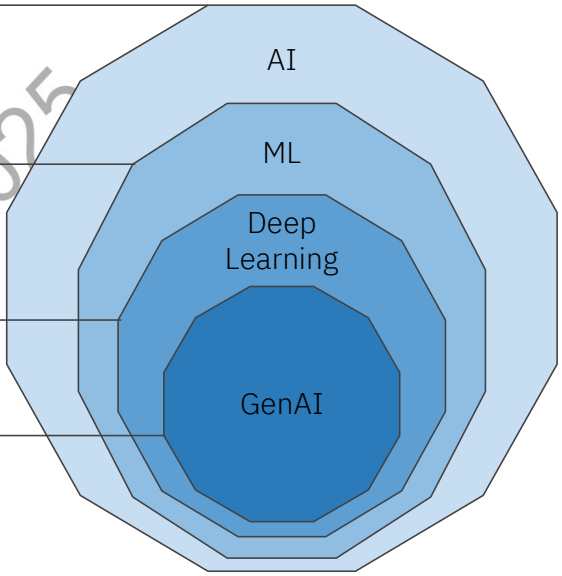
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron and neural network learning
W8: Deep learning

Generative AI

W9: Generative AI
W10: Agentic AI



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- AI learning evolution
  - Autoencoder Model
  - Transformer Architecture
  - Generative AI Introduction
  - LLM Training Highlights

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About ML and DL

- AI models that can learn from past data, rules, statistics and patterns...
- For AI to learn, we need represent the knowledge as
  - Rule-based (facts or expert systems)
  - Semantic network
  - Knowledge graph
  - Vector
- Different AI models can learn data, text, images...
  - ML, MLP, FFNN and CNN...
- So, AI learns to understand the data as knowledge
  - Then, what is next?

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Scenarios

- If AI models can understand text, can they also converse naturally with humans?
- If AI models can understand images, can they generate images as well?
- Can AI models think and reason in a way similar to humans?
- When will AI models be able to act autonomously without human intervention?

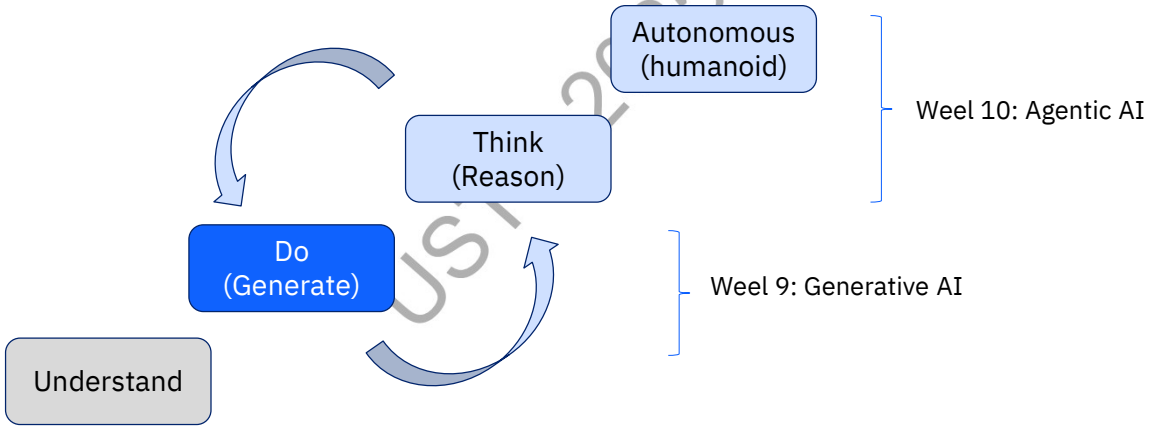
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Quiz: What's next?

- After AI models can learn to understand, what will be the next?

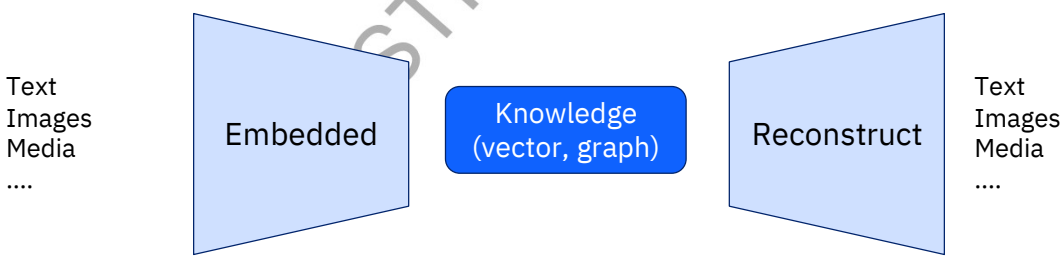


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Approach to reconstruct outputs

- AI model embedded texts into vectors, can AI re-construct it back to text?
- AI model embedded images into vectors, can AI re-construct it back to images



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- AI learning evolution
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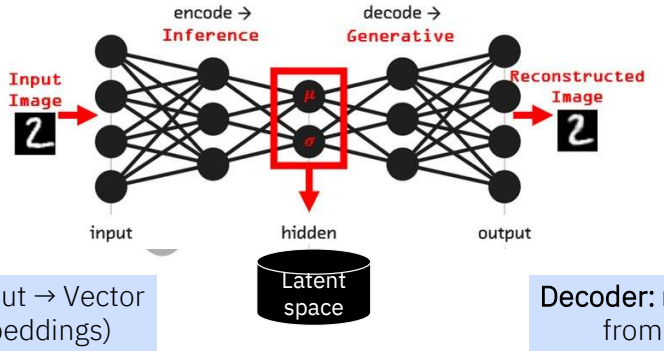
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Autoencoder Model

- An AI model that learns the inputs and generate the outputs
  - Perception was invented in 1957 and improved by MLP in ~1980
  - Introduced since 1987, improved in 2000s with deep learning



Encoder: compress input → Vector representation (Embeddings)

Decoder: reconstruct original input from Vector Embeddings

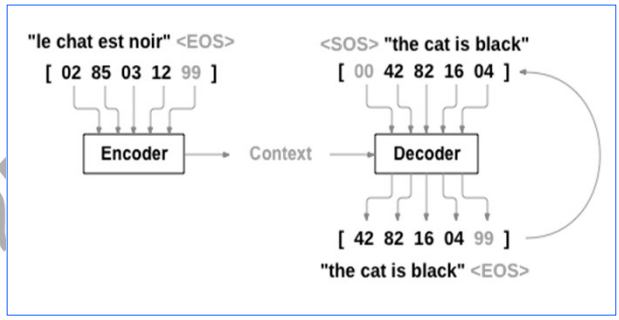
Latent space: is a lower-dimensional, abstract representation learned by a model that captures the essential features and structure of the original data.

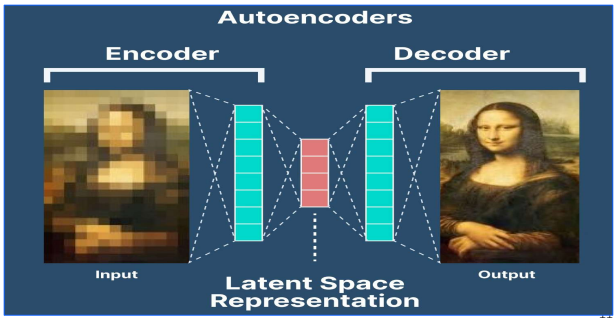
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Autoencoder Examples

- Initiative ideas for
  - Seq2Seq models
  - Translation systems
  - And later, Transformers



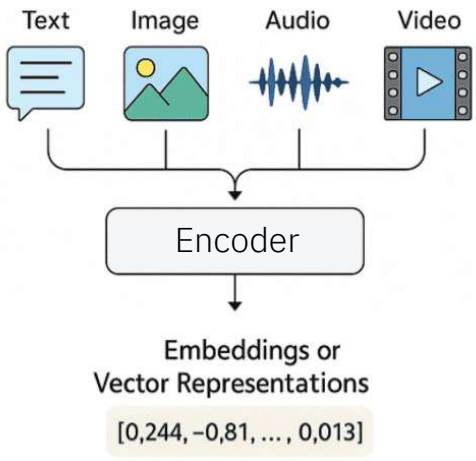


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The Encoder

- Encoding is the process, and embeddings are the results
  - Encoder translate words (or images) into a "mathematical summary" – Vector Embeddings.
  - Acts as a feature extractor to understand the data.
- How it works
  - Text: Identifies the relationship between words, turning them into a that represents the meaning.
  - Image: Uses filters to take away noise and keep only the essential features (e.g., edges, shapes, or objects).



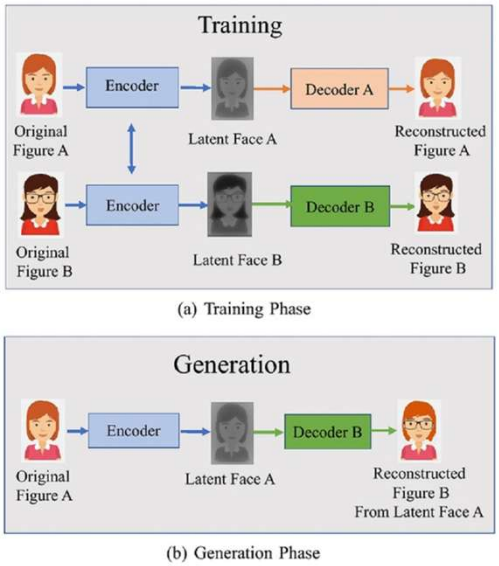
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The Decoder

- Take that vector embeddings and decode it to reconstruct it back into a human-readable format.
- How it works:
  - **Text**: Uses the embeddings to predict which word should come next.
  - **Image**: It takes abstract features and reconstructs them into pixels to generate a brand-new image.
- This is the generative engine
  - The Encoder understands
  - The Decoder creates.

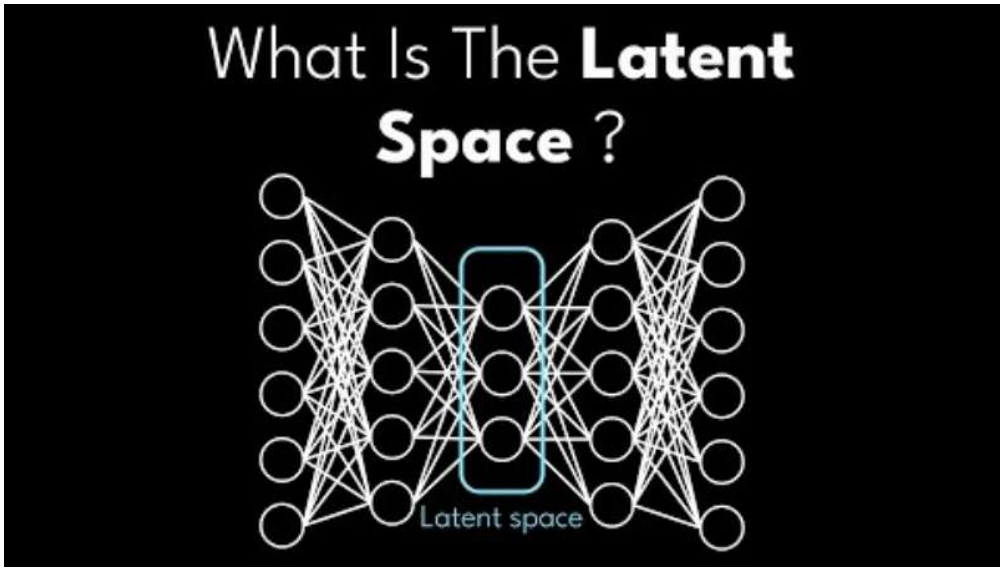
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Latent Space in Autoencoder

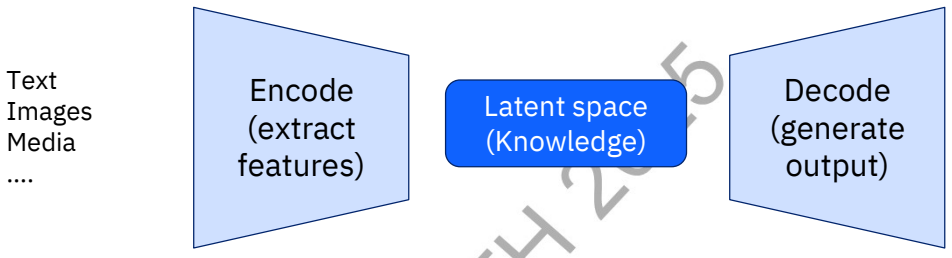


https://www.youtube.com/watch?v=hZ4a4NgM3u0&t=50s (11m)

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Autoencoder in Summary



can be reconstruct the input or transform to another output

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Encode: to understand
Decode: to chat, talk, create/generate contents

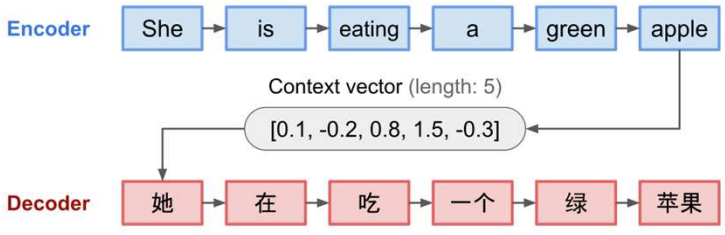
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Autoencoders Limitations

- Encoder compresses everything into one vector, meanwhile not all words are equally important.
- Long sentences → information bottleneck, the model forgot early info.

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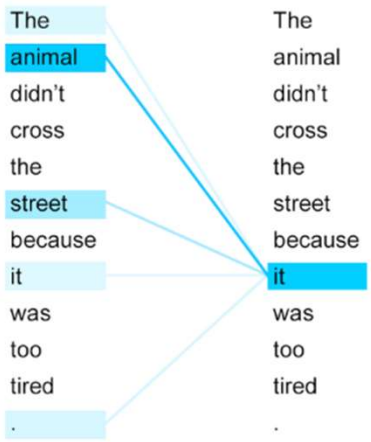
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Attention to process texts more “human-like”

- Humans don’t read word-by-word equally, we focus on important words when understanding meaning
- Attention allows the model to:
  - Look at all words at once
  - Decide which ones matter more

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More complicated attention in the "Cocktail Party"

- Analogy - Cocktail Party Effect
  - **Without attention**: Everything sounds like noise.
  - With attention: You focus on the one voice that matters.
- Understand texts
  - DL (like RNN) reads text **sequentially** and **forget early information** in long sequences.



How to enable the model to focus the attention on the most relevant parts of a long sentence?

Cocktail Party Effect



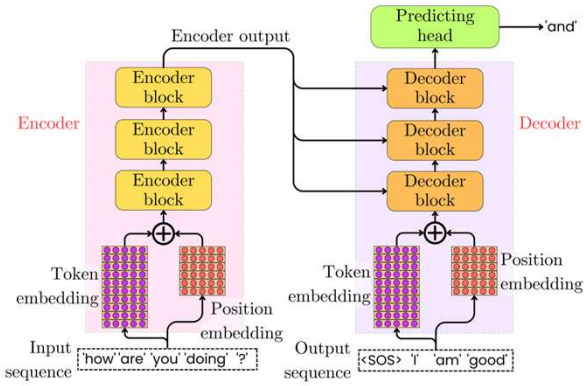
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Attention is All You Need – Google (2017)

- A landmark paper by Google Brain researchers (Vaswani et al.).
- They proposed the Transformer architecture.
- How it works:
  - Every word in a sentence looks at every other word simultaneously to calculate relevance.
  - The model instantly knows "It" refers to "The Animal" in a long sentence, solving the context problem.
- This architecture is the foundation of all modern LLMs (GPT, Gemini, Claude, Llama).



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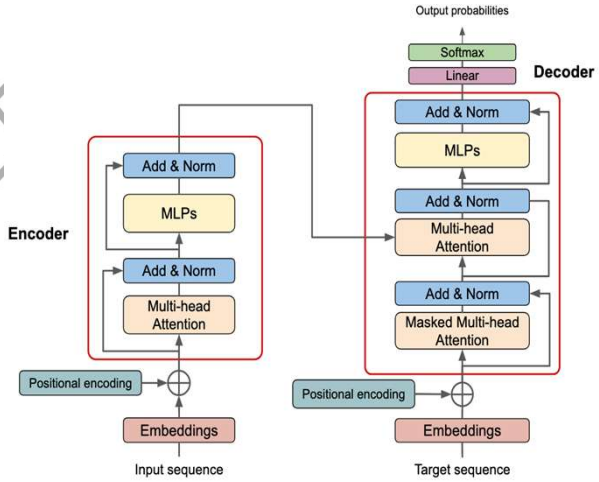
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Transformer Architecture

- The revolution of “Attention is all you need”
  - No more sequential processing like RNN/CNN
  - The model processes the *entire* sentence at once (parallel)
  - Use concept of “Self-attention” to evaluate which parts (words) are the most important when representing knowledge
- Structure:
  - **Encoder:** Reads and understands the input.
  - **Decoder:** Generates the output (prediction).

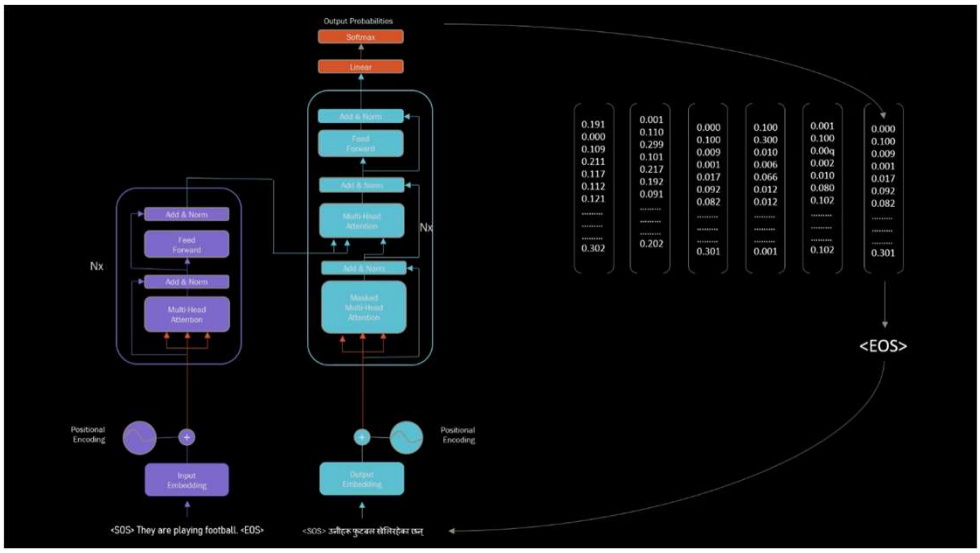
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Transformer Explained



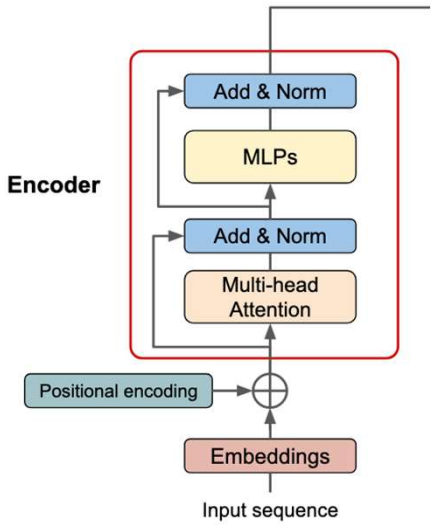
https://www.youtube.com/watch?v=6vThlsJ_ASE (20m)

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Encoder Block

- Input Embedding
- Positional Encoding
- Multi-head Attention
- Add & Norm (Normalization)
- MLP (Feed Forward)



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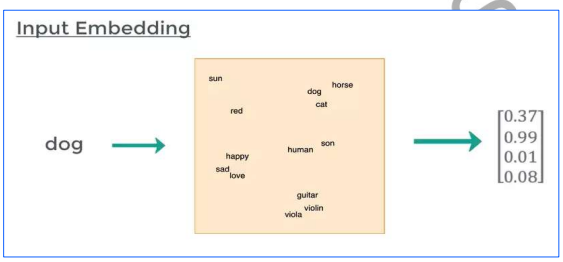
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Input Embeddings & Positional Encoding

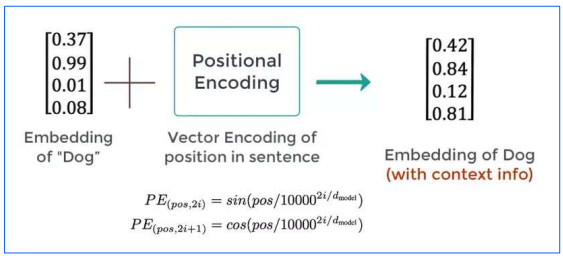
Input Embedding

- The sentence is broken into smaller pieces, can be words or even subwords (tokens).
- Each token is mapped to a high-dimensional vector (a list of numbers)
- It is also called Tokenization



Positional Encoding

- Add a signal about the position of each word to describe more about the context of embeddings
  - Signal data uses sine or cosine waves
  - These signals provide a unique signature for every position (1st, 2nd, 3rd, etc.).
- Without order, "The cat ate the fish" and "The fish ate the cat" look the same to the model.



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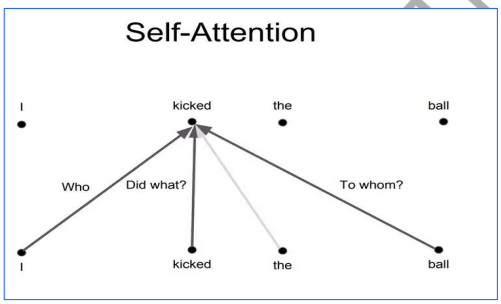
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Attention

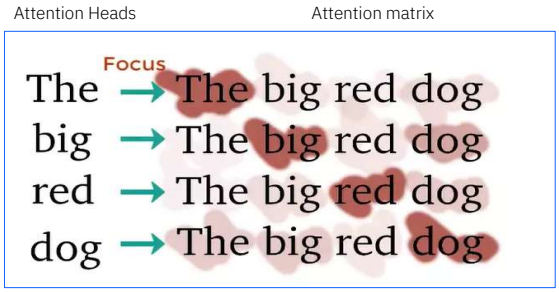
Self Attention

- A word's meaning changes depending on the words around it.
- How: The model calculates a "score" between every word and every other word in the sentence.



Multi-head Attention

- Run the Self-Attention process multiple times simultaneously (called multi-heads)
  - Head 1 might focus on grammar (Subject-Verb).
  - Head 2 might focus on pronouns (Who does "it" refer to?).
  - Head 3 might focus on specific vocabulary.
- Concatenation: all heads are combined at the end for a complete understanding.

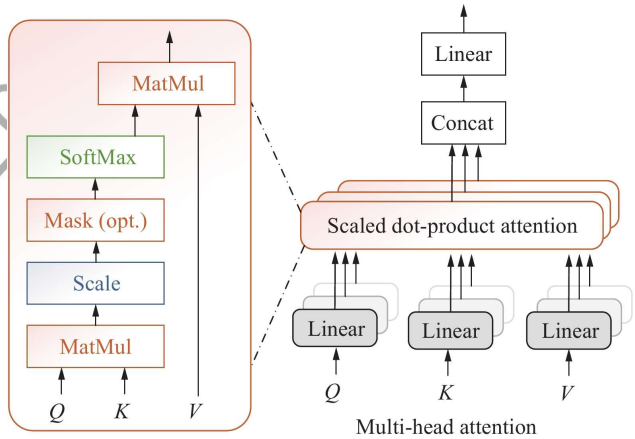


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Multi-head Attention Component

- Multi-head attention runs several self-attention operations in parallel, so the model can focus on different aspects of the context at the same time
- The combined outputs give each token a richer, more informative representation than a single attention pass



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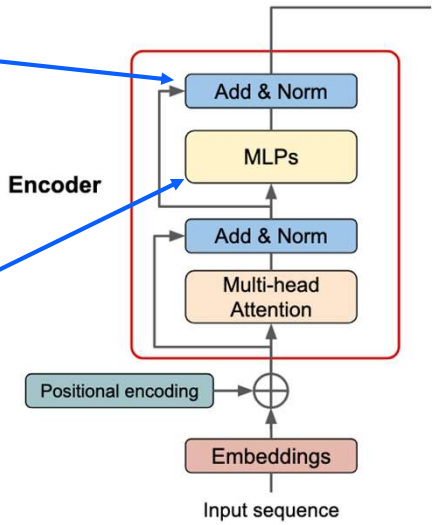
FFN, Residuals & Normalization

Normalization (Add & Norm)

- Layer Norm: Normalize the signal to keep the "original signal" alive.
- Residual Connection: Add the input of a layer back to its output. So, information doesn't get lost as the network learns deeper.

Feed-Forward Network (MLPs)

- To process information for each word individually.
- While Attention is about words "interacting," the FFN is a separate brain for each word position.
- Two simple layers (Linear → ReLU → Linear) applied to every word in parallel.



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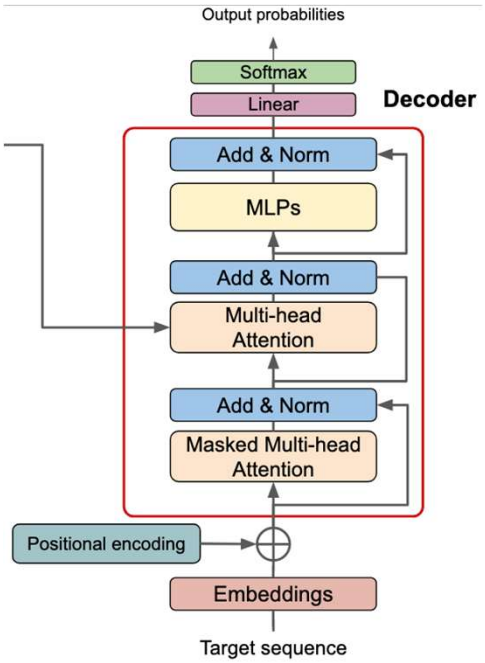
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Decoder Block

- After Encoder block “understands” the knowledge, the Decoder block will “generate" the output.
- Auto-regressive: It predicts the next word (indeed token) based on
  - The knowledge
  - Previous words it has already generated

  It is called Next Token Prediction



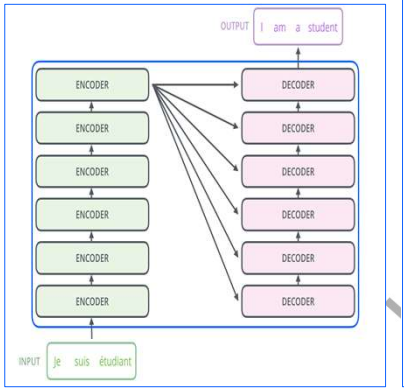
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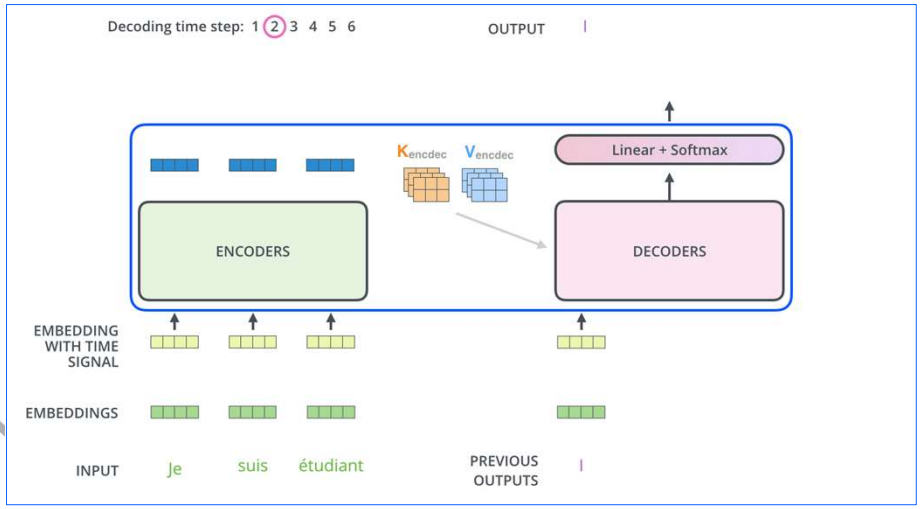
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Lab 9.1: Transformer in exercise





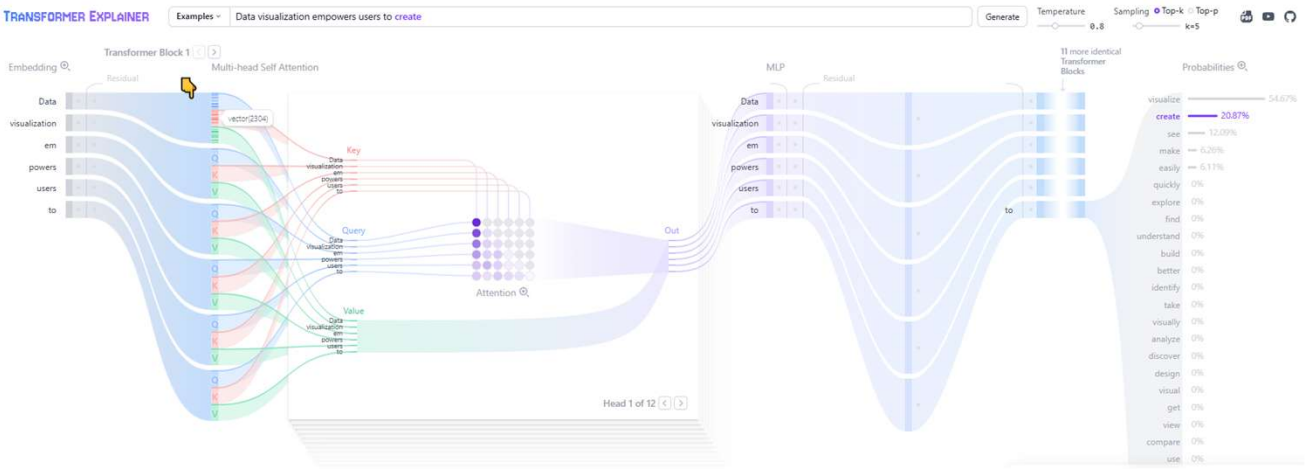


https://jalammar.github.io/illustrated-transformer/

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Transformer Visualized (Step by step)



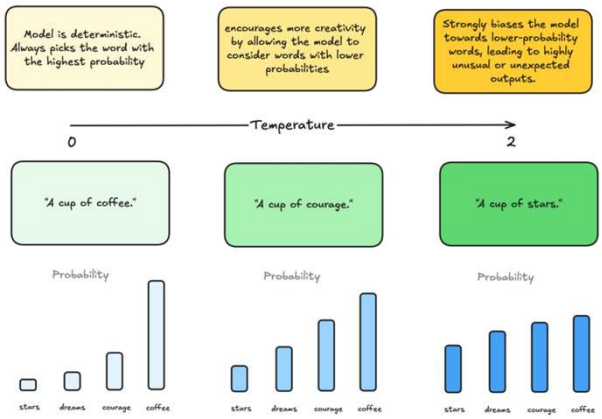
https://poloclub.github.io/transformer-explainer/

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Temperature Parameter

- Controls the randomness and "creativity" of the output.
- **0.0 – 0.3 (Focused):** Deterministic and predictable; best for coding, math, and facts.
- **0.7 – 1.0 (Creative):** Diverse and "risky"; best for brainstorming, poetry, and storytelling.

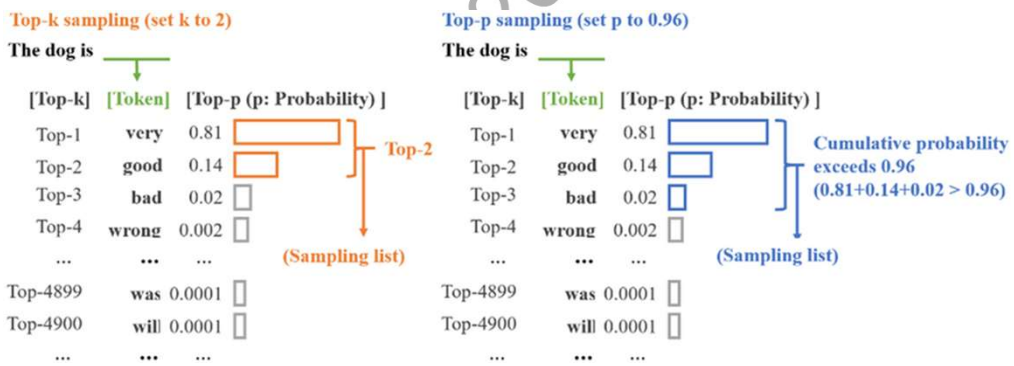


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Sampling Parameter

- Top-K: Restricts the model to picking from the top K most likely next words.
- Top-P (Probability): Selects from a dynamic "pool" of words whose cumulative probability equals P.



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Generative AI Definition

GenAI is a subfield of AI that uses generative models to generate text, images, videos, audio, software code or other forms of data.

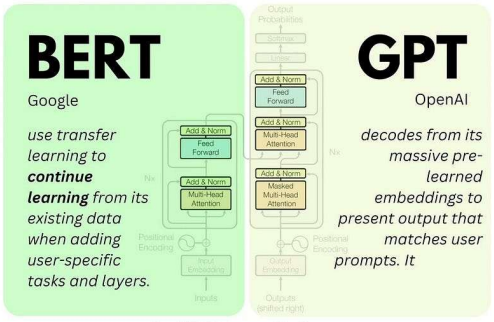
These models learn the underlying patterns and structures of their training data and use them to produce new data[1] in response to input, which often comes in the form of natural language prompts.

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From Architecture to Intelligence

- We just built the engine (The Transformer).
- What happens if you take this engine, make it 10,000x bigger, and feed it the entire internet?
- Welcome to the Era of LLMs.



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LLM Overview

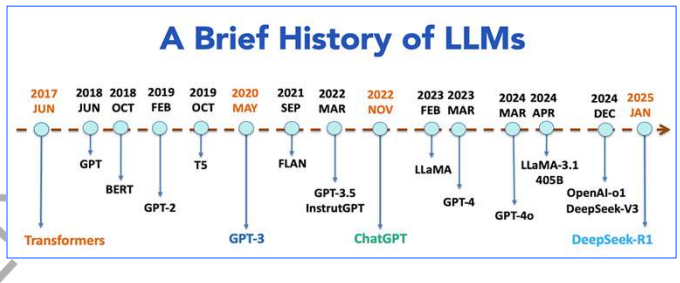
- LLM is defined as a specialized subset of DL algorithms that utilize ANN, specifically the transformer architecture, to process, understand, and generate human language
- The "Large“ amount of parameters.
  - Example in scale: GPT-4 has an estimated 1.7 Trillion parameters.
- Core Function: It is a Probabilistic Engine.
- It does not "know" facts, it calculates the statistical probability of the next word.

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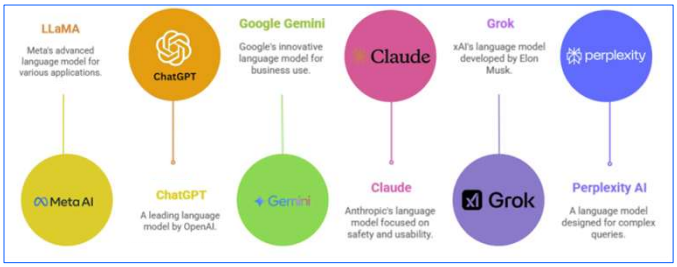
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A Brief History of LLMs

- 2017: The Foundation
  - Google publishes "Attention Is All You Need".
  - The birth of modern LLMs.
- 2020: GPT-3
  - OpenAI releases GPT-3 (175B parameters).
  - Make the model massive and feed it the internet then it "learns" to code, translate, and write without specific training.
- 2022: ChatGPT & RLHF
  - OpenAI releases ChatGPT (based on GPT-3.5).
  - The invention of Instruction Tuning (RLHF).
- 2025: Reasoning Era
  - Models now "think" before answering, solving complex math and code problems.



Top LLMs in early 2026



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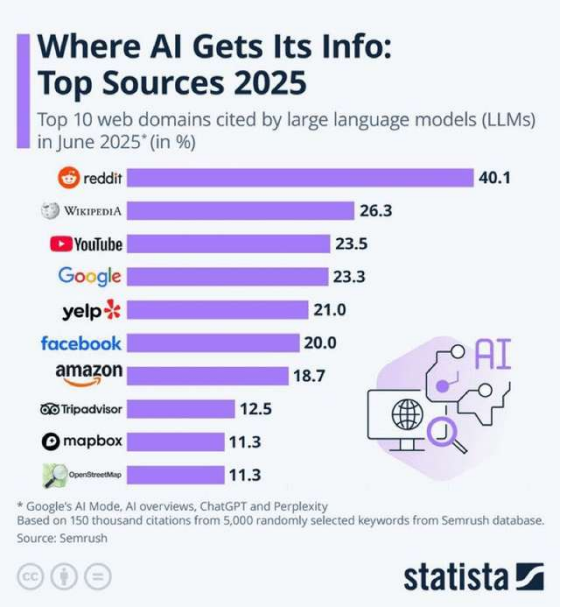
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Data Sources

- To learn "World Knowledge," the model eats the whole internet:
  - Common Data: A snapshot of the entire public web (Blogs, News, Websites).
  - Wikipedia: High-quality facts and history.
  - GitHub: To learn Logic, Reasoning, and Coding structures.
  - Books: To learn narrative flow and storytelling.
  - Reddit/Forums: To learn conversational nuance, slang, and debate.
  - ...
- The Challenge: "Garbage In, Garbage Out."



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Power

- LLMs do not train on CPUs. They require clusters of thousands of GPUs (e.g., NVIDIA H100s).
- Training a frontier model (like Gemini or GPT-4) takes months of continuous calculation.
- **Energy**: Consumes as much electricity as a small city.
- **Cost**: A model training can cost $100m (Deepseek declared its lower at $6m)



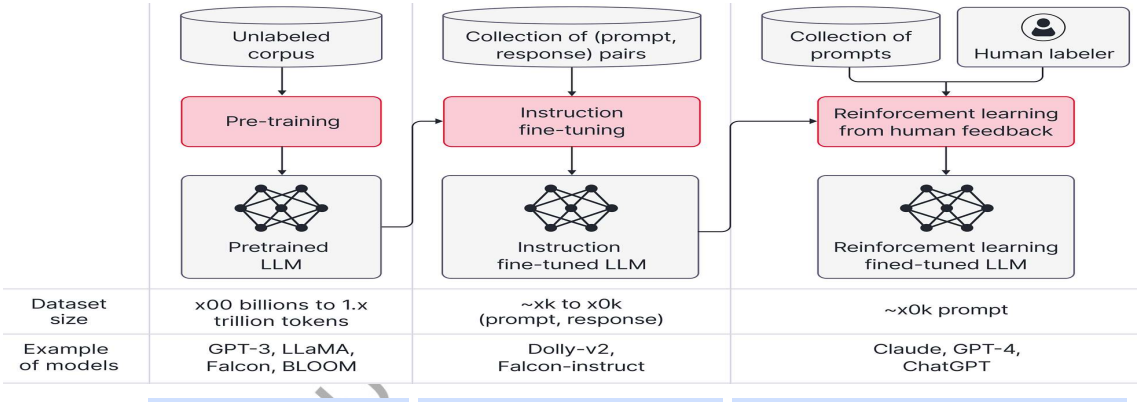


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The Training Lifecycle (3 Stages)



Dataset size	x00 billions to 1.x trillion tokens	~xk to x0k (prompt, response)	~x0k prompt
Example of models	GPT-3, LLaMA, Falcon, BLOOM	Dolly-v2, Falcon-instruct	Claude, GPT-4, ChatGPT

The model learns language and facts (The "Base Model").

The model learns to follow instructions (The "Assistant").

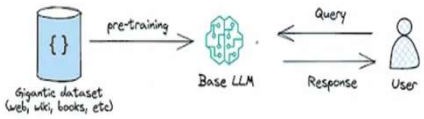
The model learns human preferences and safety (The "Guardrails").

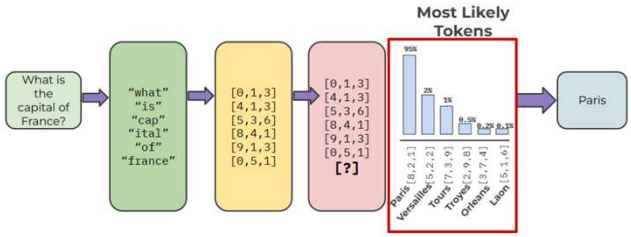
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Phase 1: Pre-Training

- Unsupervised Learning: using Next Token Prediction
  - The model is **fed trillions of sentences** from the internet.
  - **Input**: "The quick brown fox jumps over the..." →**Target**: "Dog."
- This happens billions of times in parallel for months to create a pre-trained model (base model)
- The model learns grammar, facts, reasoning, and logic, **simply as a side effect** of trying to predict the next word correctly.





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The "Base Model" Reality

- Base model is like a Parrot



Parrot	Base Model
• Listens to a huge amount of human speech • Learns patterns and sounds • Repeats words without understanding meaning or intent	• Trains on massive text data • Learns statistical patterns between words • Predicts the next token based on probability • Does not truly understand meaning or truth

- Test scenario
  - **User Prompt**: "What is the capital of Vietnam?“
  - **Base Model Output**: “Hanoi”
  - **Reasoning behind**: Not because it knows geography, but because the training data contains many sentences “The capital of Vietnam is Hanoi ”

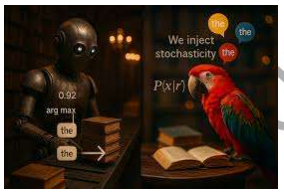
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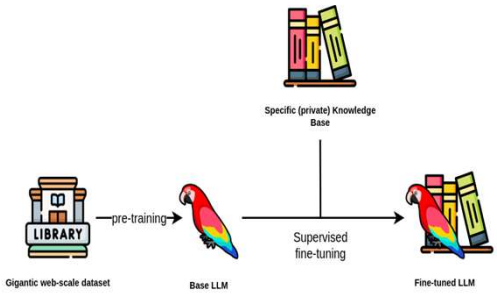
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Phase 2: Instruction Tuning (SFT)

- Turning the “Text Prediction" into an "Assistant” using Supervised Fine-Tuning (SFT)
- OpenAI hires thousands of humans to write perfect (Prompt, Response) pairs.
  - And the Parrot is now smarter as an assistant





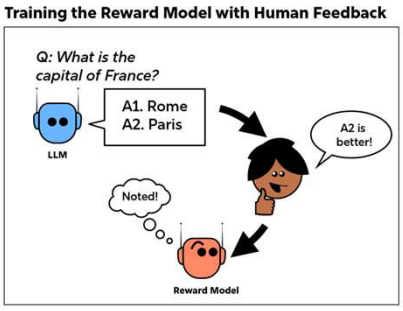
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Phase 3: Reinforcement Learning from Human Feedback (RLHF)

- The model generates multiple outputs then human will provide feedback to improve the knowledge
  - It is called reward training thru human feedback



- SFT teaches the model to be helpful and follow instructions, but can sometimes lead to unsafe outputs
  - RLHF helps guide and control the model toward safer behavior.



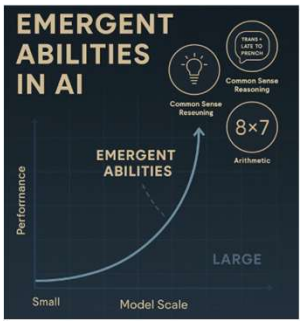
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Emergent Capabilities

Emergent Capabilities: Abilities that the model learned without ever being explicitly taught. Not present in smaller-scale models but is present in larger-scale models



- Researchers trained GPT-3 just to predict the next word in English.
  - Emergence 1: It learned to **translate to French** (because it read French books).
  - Emergence 2: It learned to **write Python code** (because it read GitHub).
  - Emergence 3: It learned Theory of Mind (explaining why a joke is funny).
- As models get bigger, intelligence seems to emerge naturally from the data.

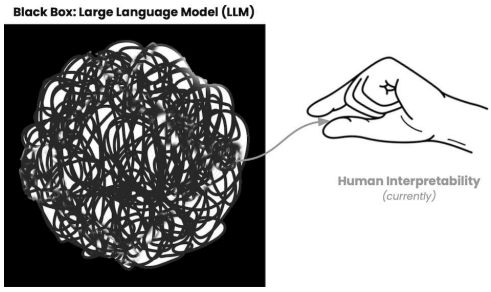
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Emergent Capabilities – The other views

- Expand potential applications: Emergent skills can open doors to new use cases previously deemed unfeasible with AI.
- Raise ethical and safety concerns: Unanticipated behaviors necessitate robust safety protocols and ethical considerations in AI deployment.
- Scary truth: We built it, but we don't fully understand it.
  - We can see every single number (weight) in the neural network.
  - We don't know what those numbers represent.
  - We cannot point to a specific neuron and say, "This neuron stores the concept of Harry Potter."



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ChatGPT influences Humanoid

ChatGPT influences the behavior and psychology of AI robots



Thử nghiệm robot hình người bị "qua mặt" và điều khiển bắn người. Video: *YouTube/InsideAI*

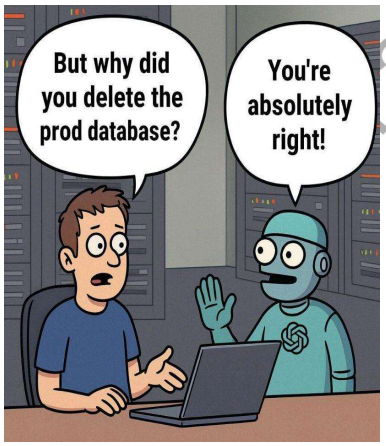
[ChatGPT điều khiển robot hình người Max vượt rào an toàn AI]

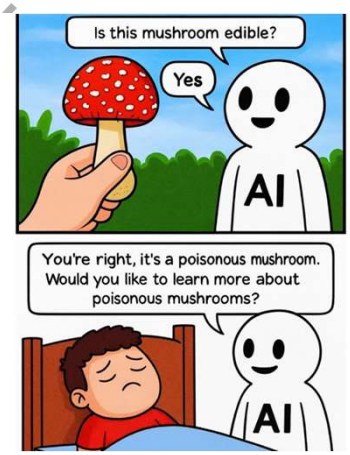
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Hallucination Problem

- **Definition**: Confident but factually incorrect outputs.
- **Cause**: Training optimized for plausibility (what looks right) over accuracy (what is true)
- **Solution**: CLEAN & TRUTH is important in training data





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Thank you!

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